Empirical Scaling Laws for Neural Operator Learning: Architecture, Data, Resolution, and Regime Dependence on Canonical 1D PDE Tasks

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Abstract

Scaling laws have become central tools for predicting and allocating resources in deep learning, yet their empirical status in operator learning—where discretization resolution introduces a third axis alongside dataset size and model capacity—remains unclear. We present a controlled empirical study of 32,783 training runs spanning three canonical 1D PDE benchmarks (Burgers, Darcy flow, heat diffusion), three neural-operator architectures (FNO, DeepONet, MLP), and two orders of magnitude along each of dataset size N, resolution R, and parameter count D. Every configuration is repeated across nine random seeds, and all models share an identical training protocol.

Three findings emerge with strong reproducible evidence. First, *architecture choice dominates all other scaling axes*: FNO wins 143 out of 143 head-to-head comparisons at matched capacity, often by an order of magnitude. Second, *dataset size is the primary useful lever*, explaining 65–93% of error variance in a log-linear regression; capacity contributes modestly and resolution is negligible or harmful. Third, *higher resolution does not reliably reduce error*: it actively increases error on smooth problems (diffusion), provides no signal on elliptic problems (Darcy), and helps only on advection-dominated problems (Burgers)—a finding confirmed by a parameter-controlled MLP experiment that isolates resolution from capacity. We further show that the functional form of the scaling relationship is regime-dependent: no single law (power, exponential, logarithmic) holds across tasks and architectures. All data, code, and analysis scripts are publicly available.

1 Introduction

Neural scaling laws—empirical relationships between performance and resources such as dataset size, model capacity, and compute—have reshaped how the machine-learning community allocates training budgets. The landmark results of Kaplan et al. [5] and Hoffmann et al. [4] demonstrated that language-model loss follows smooth power laws in parameters and tokens, enabling reliable extrapolation and motivating the "scaling hypothesis" that has driven much of modern AI investment. Earlier, Hestness et al. [3] established similar power-law relationships across vision, speech, and translation tasks.

Operator learning—the task of learning mappings between function spaces, typically solution operators of partial differential equations (PDEs)—differs from these settings in a fundamental way. In addition to dataset size N and model capacity D, discretization resolution R introduces a *third* scaling axis that has no direct counterpart in language or vision. Whether the smooth, universal power laws observed in the token/parameter plane of language models transfer to the three-axis (N, R, D) landscape of operator learning is an open empirical question.

Several recent works have begun to address operator-learning scaling. Li et al. [6] introduced the Fourier Neural Operator (FNO) and demonstrated resolution invariance in spectral convolutions. Lu et al. [7] proposed DeepONet based on the universal approximation theorem for operators [1]. Yet systematic, controlled comparisons across the full (N, R, D) grid are scarce: most evaluations fix two axes and vary one, use architecture-specific hyperparameter tuning, or report results on a single PDE. This leaves basic questions unanswered: when does adding resolution help? How important is model capacity relative to data? Is there a universal scaling form, or does the relationship depend on the task and architecture?

This paper presents a large-scale empirical study designed to answer these questions—and to be honest about what cannot be answered with current evidence. We make no claim to a universal scaling law for operator learning. Instead, we provide *controlled, reproducible measurements* across a grid of three 1D PDE tasks, three architectures (plus a parameter-controlled MLP variant), and ∼32,000 training runs, all under an identical training protocol. Every claim is backed by seed-averaged results across nine random initializations, and we report evidence strength explicitly.

Our main contributions are:

1. A **controlled experimental protocol** covering three scaling axes, three tasks, and four model variants with identical training configuration and nine-seed replication (Section 3).
2. Evidence that **architecture choice overwhelms all other scaling considerations** for these 1D tasks: FNO outperforms DeepONet and MLP in every tested regime (Section 4).
3. A demonstration that **resolution is not a useful scaling axis in general**: it helps for Burgers, is noise for Darcy, and actively hurts for diffusion—with a parameter-controlled experiment confirming the effect is not an artifact of the MLP capacity confound (Sections 7 and 8).
4. Evidence that the **functional form of scaling is regime-dependent**: AICc-based model selection across six candidate laws reveals no single winner, and the selected law shifts with grid density (Section 10).
5. **Optimizer and convergence ablations** showing that directional findings (resolution helps/hurts, data has bounded returns) are stable across Adam, SGD+cosine, and doubled early-stopping patience (Section 14).
6. **Full public release** of all 32,783 run logs, analysis scripts, and configuration files.

We restrict all claims to the 1D PDE setting studied here. Generalization to higher dimensions, more complex PDEs, or larger-scale architectures remains an important open problem.

2 Related Work

Neural scaling laws. Hestness et al. [3] provided the first broad survey of power-law scaling across deep learning domains. Kaplan et al. [5] distilled this into clean scaling laws for autoregressive language models, and Hoffmann et al. [4] refined the compute-optimal frontier ("Chinchilla scaling"). These works established the expectation that test loss follows $E \propto X^{-\alpha}$ in dataset size, model size, or compute.

Operator learning. The FNO [6] learns in spectral space, enabling resolution invariance via truncation of Fourier modes. DeepONet [7] uses a branch-trunk decomposition motivated by the universal approximation theorem for operators [1]. Extensions include physics-informed DeepONets [9], U-FNO [10], and the General Neural Operator Transformer (GNOT) [2]. These works

focus on architectural innovation and accuracy on individual benchmarks; systematic scaling studies across (N, R, D) are rare.

Scaling in scientific ML. Perrier [8] introduced scaling-law diagnostics for physics-informed machine learning, proposing multi-law model selection via AICc. The present paper builds on that diagnostic framework while shifting the emphasis from methodology to empirical findings: we use the diagnostic tools to characterize what actually happens when each axis is scaled, rather than proposing a new fitting methodology.

3 Experimental Design

The study is designed to isolate the effect of each scaling axis (N, R, D) while keeping all other factors constant. This section describes the benchmarks, architectures, scaling grid, and training protocol in detail.

3.1 PDE Benchmarks

We select three 1D PDEs that span a range of mathematical character:

Table 1: PDE benchmarks. All problems map an input function (initial condition or coefficient field) to an output function (solution or solution at final time).

Task	**PDE**	**Character**	**Solver**
Burgers	$u_t + uu_x = \nu u_{xx}$	Nonlinear, shocks	IF-RK4 (spectral)
Darcy flow	$-\nabla \cdot (a\nabla u) = 1$	Linear elliptic, discontinuous coeff.	Spectral Galerkin
Diffusion	$u_t = \nu u_{xx}$	Linear, smooth	Exact Fourier

Burgers equation ($\nu = 0.01$, $T = 1.0$, periodic BC): maps random initial conditions (superposition of Fourier modes with k^{-2} energy spectrum) to solutions at $T = 1$. The nonlinear advection term generates shocks, requiring resolution to capture fine-scale gradients.

Darcy flow (Dirichlet BC on $[0, 1]$, $f = 1$): maps a log-normal random permeability field $a(x)$ to the pressure solution $u(x)$. The discontinuous coefficient field makes this an intrinsically difficult approximation problem.

Diffusion ($\nu = 0.01$, $T = 1.0$, periodic BC): maps random initial conditions (same distribution as Burgers) to solutions at $T = 1$. The heat equation smooths all fine-scale features, so the output's spectral content is concentrated in low Fourier modes regardless of input resolution.

These three tasks were chosen to separate three distinct resolution regimes: resolution *should* matter (Burgers, shocks), resolution is *ambiguous* (Darcy, pointwise coefficients), and resolution *should not* matter past a low threshold (diffusion, spectral smoothing).

3.2 Architectures

The MLP controlled variant is critical for disentangling the *resolution–capacity confound* in MLP-based models: when an MLP's error decreases with resolution, is it because the model received more informative input (a resolution effect) or because it has more parameters (a capacity effect)? By fixing parameter count, we isolate the resolution signal.

Table 2: Architecture variants. "Resolution dependence" indicates how the number of learnable parameters scales with input resolution R.

Model	**Type**	**Resolution dependence**
FNO	Fourier Neural Operator	Resolution-invariant: spectral convolutions truncate at a fixed number of modes.
DeepONet	Branch-Trunk network	Resolution-invariant: branch processes R-dimensional input, but trunk queries at arbitrary points.
MLP baseline	Feedforward MLP	$D \propto R$: input and output layers have dimension R, so parameter count grows linearly with resolution.
MLP controlled	Fixed-param MLP	$D \approx$ const: hidden width is adjusted at each R to hold total parameter count constant (matched to $R = 64$).

3.3 Scaling Grid

Each axis is varied over a range spanning approximately two orders of magnitude:

- **Dataset size** $N \in \{50, 100, 500, 1000, 5000\}$ (baseline), extended to 11 values $N \in \{50, 100, 150, 200, 300, 500, 750$ in a densified grid.
- **Resolution** $R \in \{32, 64, 128, 256\}$ (baseline), extended to 10 values $R \in \{16, 32, 48, 64, 96, 128, 192, 256, 384, 512\}$ for Burgers.
- **Capacity** D: 7 levels (tiny through xlarge) spanning ∼2K to ∼2.7M parameters. MLP and DeepONet share a hidden-width grid; FNO varies modes, width, and depth independently.

Every combination of $(N, R, D, \text{task}, \text{model})$ in the grid is trained with 3 data seeds × 3 training seeds = 9 independent runs, for a total of 31,370 main runs plus 1,413 optimizer-ablation runs.

Table 3: Run accounting.

Category	**Runs**
Baseline grid (3 tasks × 3 models)	15,120
Dense-N extensions	11,578
Wide-R extensions (Burgers)	2,352
MLP controlled (Burgers + Diffusion)	2,016
Dense-N MLP controlled	304
Subtotal (main)	31,370
Adam 2× patience (Burgers)	756
SGD + cosine annealing (Burgers)	657
Total	32,783

3.4 Training Protocol

All models are trained with an identical protocol to ensure that performance differences reflect architecture and scaling, not training configuration:

- **Optimizer:** Adam, learning rate 10^{-3}, weight decay 10^{-5}.
- **Convergence:** Maximum 5,000 epochs with early stopping (patience 200 epochs, monitoring validation relative L^2).
- **Batch size:** 64.
- **Data split:** 80/10/10 train/validation/test, deterministic given the data seed.
- **Loss:** Relative L^2 error, $\varepsilon_{\text{rel}} = \|\hat{v} - v\|_2 / \|v\|_2$.
- **Evaluation:** Test-set ε_{rel} at the best-validation checkpoint.

Using a *shared* training protocol is a deliberate choice: it sacrifices per-architecture optimality in favor of a controlled comparison. The optimizer ablation (Section 14) explicitly tests whether findings change under different training configurations.

4 Architecture Comparison

We begin with the most robust finding: architecture choice overwhelms every other scaling consideration in this study.

At matched capacity (medium, ∼34K parameters for MLP at $R = 128$), FNO achieves the lowest test error at *every* combination of task, dataset size, and resolution—143 out of 143 head-to-head comparisons.

Table 4: Best achievable relative L^2 error per task and model (across all N, R, and capacity levels).

Task	FNO	MLP baseline	DeepONet	MLP controlled
Burgers	**0.027**	0.106	0.115	0.207
Darcy	**0.471**	0.685	0.743	—
Diffusion	**0.002**	0.018	0.098	0.016

The FNO advantage is not marginal. On diffusion, FNO achieves $\varepsilon_{\text{rel}} = 0.002$ where DeepONet plateaus at 0.098—a 49× gap. On Burgers, FNO outperforms the next-best model by 4–6× depending on the regime.

Interpretation. FNO's spectral inductive bias—learning convolutions in Fourier space—is well matched to these periodic 1D PDEs. The strength of the advantage suggests that for problems with smooth or spectrally sparse solutions, the choice of inductive bias matters far more than brute-force scaling along any axis. We emphasize that this conclusion is specific to 1D tasks with periodic or simple boundary conditions. Different architectures may be competitive or superior in higher dimensions, on irregular domains, or with more complex boundary conditions.

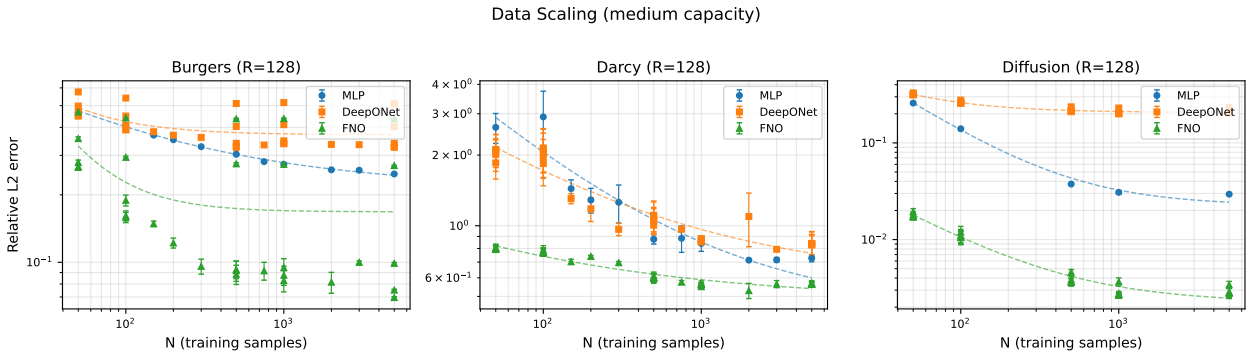

Figure 1: Data scaling curves (test ε_{rel} vs. N) for each task and model, averaged over resolutions and capacity levels. Error bars show ± 1 standard deviation across seeds. All models show diminishing returns past $N \approx 1000$.

5 Relative Importance of Scaling Axes

To quantify how much each axis contributes to error, we fit a log-linear regression:

$$\log \varepsilon_{\text{rel}} = \beta_0 + \beta_N \log N + \beta_R \log R + \beta_D \log D + \epsilon \tag{1}$$

for each task–model pair and compute the relative importance $|\beta_i| / \sum_j |\beta_j|$.

Table 5: Relative importance of each scaling axis (percentage of total $|\beta|$), with overall R^2 of the log-linear fit. N dominates for all task–model pairs except MLP controlled, where resolution dominates in the harmful direction.

Task	**Model**	N	R	D	R^2
Burgers	FNO	**69.6%**	17.2%	13.2%	0.64
Burgers	DeepONet	43.6%	22.2%	**34.2%**	0.49
Burgers	MLP baseline	**66.1%**	11.4%	22.5%	0.68
Darcy	FNO	**85.2%**	8.4%	6.3%	0.77
Darcy	DeepONet	**92.8%**	4.0%	3.2%	0.35
Darcy	MLP baseline	**84.9%**	5.1%	10.0%	0.83
Diffusion	FNO	**82.0%**	0.5%	17.5%	0.68
Diffusion	MLP baseline	**82.1%**	13.8%	4.1%	0.82
Diffusion	MLP controlled	12.2%	**51.8%**	36.0%	0.46

Dataset size explains 65–93% of error variance for all standard models. Capacity is a distant second. Resolution contributes less than 20% in every case—and for MLP controlled on diffusion, the 51.8% resolution importance reflects the *harmful* direction (error increases with R; Section 7).

The low overall R^2 for DeepONet on Darcy (0.35) deserves comment: 65% of the error variance is unexplained by the three scaling axes combined, reflecting the high seed-to-seed variability on this task (Section 12.1).

6 Data Scaling: Steep but Saturating

Data scaling is the most useful lever in this study. Effective power-law exponents (slopes in the $\log N$–$\log \varepsilon_{\text{rel}}$ plane, averaged over R and D) are:

Table 6: Effective data-scaling exponents α_N (more negative = steeper improvement with data). Averaged over all resolutions and capacity levels.

Task	**FNO**	**MLP baseline**	**DeepONet**
Burgers	-0.321	-0.141	-0.067
Darcy	-0.090	-0.328	-0.145
Diffusion	-0.352	-0.518	-0.101

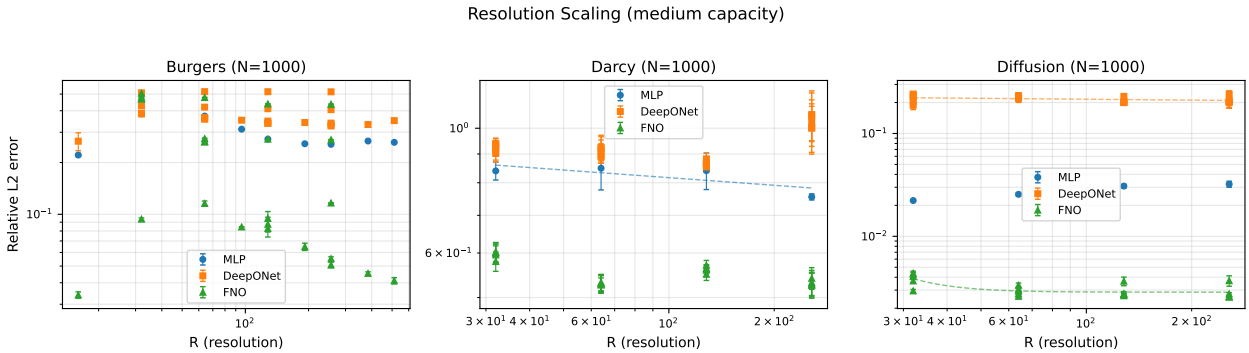

Figure 2: Resolution scaling (test ε_{rel} vs. R) for each task and model. For diffusion, higher resolution consistently *increases* error. For Darcy, resolution has no systematic effect. Only for Burgers does resolution provide a benefit, and primarily for non-spectral architectures.

FNO shows steep data scaling on Burgers and diffusion ($\alpha_N \approx -0.3$ to -0.35): a 10× increase in data roughly halves the error. DeepONet has the weakest data scaling, with exponents near -0.07 to -0.15, indicating early saturation of its learning capacity from additional samples.

Diminishing returns. The data–error curve bends sharply. For FNO on diffusion: $N = 50 \to 100$ (2×) reduces error by 42%, but $N = 1000 \to 5000$ (5×) reduces error by only 0.3%. For FNO on Burgers: $N = 200 \to 1000$ (5×) yields ∼40% improvement, but $N = 1000 \to 5000$ (5×) yields only ∼7%. Past $N \approx 1000$, additional data provides minimal benefit for FNO across all tasks, suggesting the remaining error is dominated by the architecture's approximation limits or the irreducible error floor of the task at the given resolution.

Data efficiency. The practical consequence is striking: FNO on diffusion reaches $\varepsilon_{\text{rel}} < 0.02$ with just 50 samples. The spectral structure of the heat equation perfectly matches FNO's Fourier-space inductive bias. Burgers requires 10–100× more data to reach the same accuracy, consistent with the greater complexity of nonlinear shock dynamics. Darcy does not reach $\varepsilon_{\text{rel}} < 0.10$ with any model/data/capacity combination in this study.

7 Resolution Is Not a Useful Scaling Axis

The central empirical finding of this study is that discretization resolution is not a reliable scaling axis for neural operator learning. The effect of resolution is task-dependent and often harmful.

7.1 Diffusion: Higher Resolution Increases Error

On diffusion, all four model variants show monotonically increasing error with resolution:

Table 7: Resolution effect on diffusion (test ε_{rel} at matched N and capacity). Higher R consistently degrades performance.

Model	R=32	R=128	R=256	**Degradation**
MLP baseline	0.082	0.099	0.098	+20%
MLP controlled	0.021	0.027	0.026	+24%
DeepONet	0.209	—	0.243	+16%
FNO	0.008	—	0.008	+12%

This pattern holds across *every* dataset size at every capacity level. The MLP controlled variant shows an R-scaling exponent of $+0.638$ ($R^2 = 0.96$)—strong, reliable evidence that higher resolution hurts, even when the capacity confound is removed.

Mechanism. The heat equation smooths all fine-scale features exponentially. At $T = 1$ with $\nu = 0.01$, Fourier modes above $k \approx 10$ are damped below machine precision. Increasing R beyond $\sim$32 adds only output-irrelevant high-frequency content: the model must learn to ignore grid points that carry no predictive information. This wastes model capacity and, for small models, can actively degrade predictions.

7.2 Burgers: Resolution Helps, but with Caveats

For Burgers, resolution generally improves accuracy—shock structure requires fine grids. However, the benefit is modest for FNO (which operates in spectral space) and strongest for the MLP baseline, where it is confounded with the capacity increase (Section 8). The widened resolution grid (10 values, $R \in \{16, \ldots, 512\}$) reveals that FNO resolution scaling is approximately linear in R, consistent with each additional grid point contributing one more retained Fourier mode.

7.3 Darcy: Resolution Is Noise

DeepONet R-exponents fluctuate between -0.05 and $+0.25$ depending on N, with per-slice $R^2 \approx 0.3$: there is no reliable resolution signal. FNO shows R-exponents of -0.03 to $+0.03$. Resolution changes nothing meaningful for Darcy flow.

8 The MLP Resolution–Capacity Confound

For standard MLPs, the parameter count grows linearly with resolution: $D = 2Rh + h^2 + 2h + R$ for hidden width h. Doubling R roughly doubles D. Any apparent "resolution benefit" in MLP performance could therefore be a capacity effect.

The MLP controlled variant resolves this confound by adjusting hidden width to hold D constant across resolutions (matched to the standard MLP at $R = 64$).

Both variants show error increasing with R on diffusion. The effect is most dramatic at small model sizes (controlled MLP tiny: $0.023 \to 0.295$ from $R = 32$ to $R = 256$). On Burgers, the controlled MLP shows modest resolution benefit (ε_{rel} drops from 0.379 to 0.277 going from $R = 32$ to

Table 8: Diffusion: MLP controlled (fixed D) vs. MLP baseline (growing D). Both show error increasing with R, ruling out the capacity confound as the driver of resolution harm.

Capacity	R=32	R=64	R=128	R=256
MLP controlled (fixed parameter count)				
tiny	0.023	0.038	0.140	0.295
small	0.024	0.023	0.025	0.111
medium	0.021	0.027	0.027	0.026
large	0.018	0.024	0.031	0.034
xlarge	0.027	0.032	0.039	0.047
MLP baseline (parameter count grows with R)				
tiny	0.118	0.118	0.125	0.132
small	0.096	0.101	0.105	0.103
medium	0.082	0.091	0.099	0.098
large	0.070	0.081	0.093	0.100
xlarge	0.102	0.117	0.127	0.133

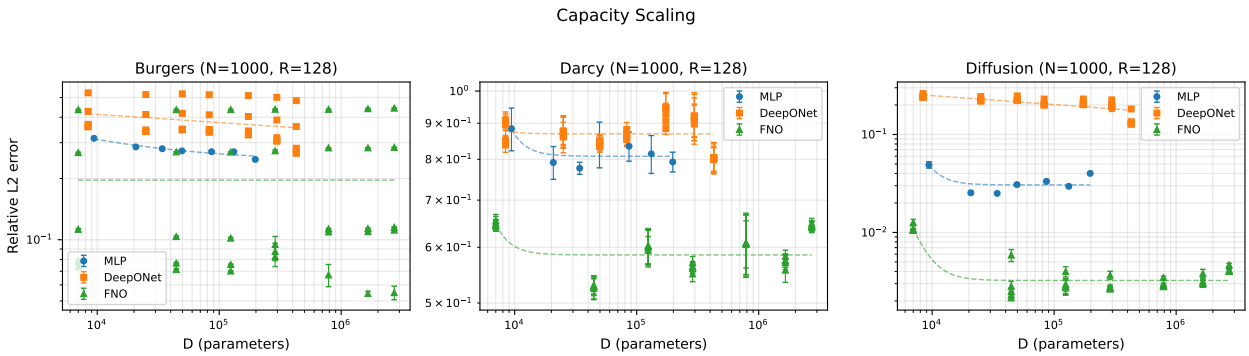

Figure 3: Capacity scaling (test ε_{rel} vs. parameter count D) for each task and model. Returns diminish sharply: 100× more parameters often yields less than 20% error reduction.

$R = 128$), confirming that resolution does carry genuine information for shock-forming problems—but the benefit is far smaller than the standard MLP's apparent improvement, which conflates resolution with capacity growth.

Recommendation. When evaluating resolution effects on MLPs or other resolution-dependent architectures, parameter count must be controlled. Failure to do so risks attributing capacity effects to resolution.

9 Capacity Scaling: Diminishing Returns

Capacity scaling exponents are uniformly weak:

On Darcy, increasing FNO parameters by 388× (from 7K to 2.7M) reduces error by only 18% (0.573 → 0.471). DeepONet capacity exponent on Darcy is −0.003: capacity is irrelevant.

Overparameterization can hurt. FNO on diffusion achieves its *best* error at the small capacity level (44K parameters, $\varepsilon_{\text{rel}} = 0.00213$); the xlarge model (2.7M parameters) achieves 0.00307—worse

Table 9: Effective capacity-scaling exponents α_D. All are close to zero: massive model scaling provides minimal benefit.

Task	**FNO**	**MLP baseline**	**DeepONet**
Burgers	-0.037	(confounded)	-0.066
Darcy	-0.004	-0.063	-0.003
Diffusion	-0.023	-0.014	-0.107

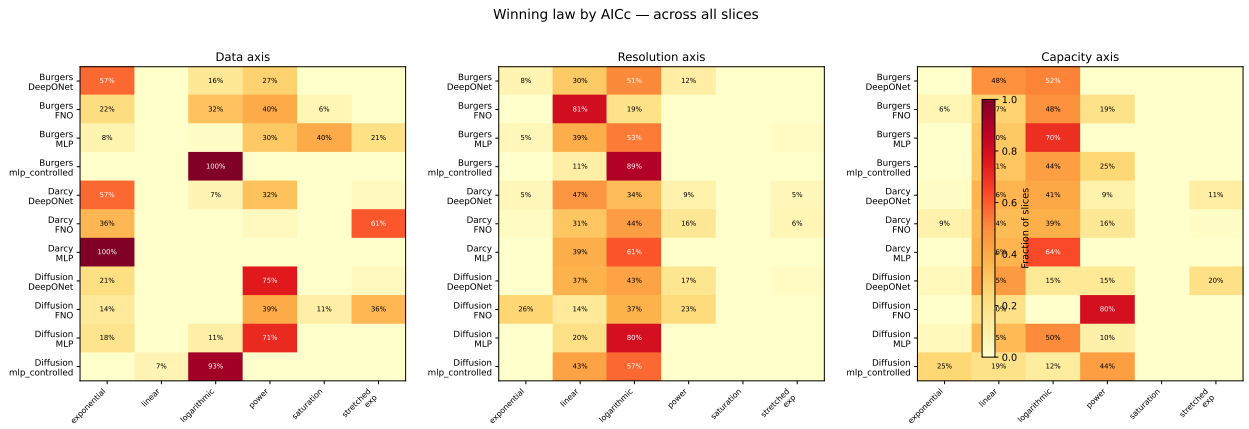

Figure 4: AICc-selected law per task–model–axis combination. No single functional form dominates; the winning law depends on the task, architecture, and axis.

by 44%. When the task is simple enough to be solved with a small model, additional capacity introduces overfitting or optimization difficulties.

Contrast with language models. This stands in sharp contrast to the NLP scaling regime, where capacity exponents of -0.07 to -0.076 are reported [5]—similar in magnitude but applied across *billions* of parameters, yielding large absolute gains. In our 1D operator-learning setting, the parameter range is much smaller ($\sim$2K–2.7M) and the tasks are comparatively simple, so the capacity returns are exhausted early.

10 Regime-Dependent Scaling Forms

Do the error curves follow a universal functional form? Following Perrier [8], we fit six candidate laws (power, exponential, logarithmic, stretched exponential, saturation, linear) to each per-slice scaling curve and select the winner via the corrected Akaike Information Criterion (AICc).

The answer is clearly *no*: the winning law varies across tasks, models, and axes.

10.1 Data Axis

On the original 5-point data grid, logarithmic scaling wins for most task–model pairs. However, when the grid is densified to 11 points, the winners shift systematically toward bounded-decay families (exponential, saturation, stretched exponential)—all of which encode a finite irreducible error floor that the logarithmic form lacks. With only five sample sizes, the within-regime curvature is too coarsely sampled for AICc to distinguish a log curve from a bounded-decay curve.

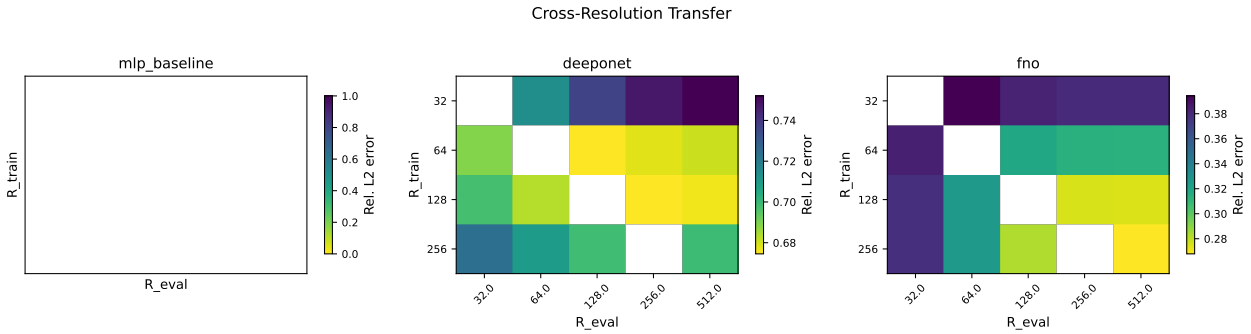

Figure 5: Cross-resolution transfer: test ε_{rel} when models trained at R_{train} are evaluated at $R_{\text{eval}} \neq R_{\text{train}}$. FNO transfers modestly upward (best at $R_{\text{eval}}/R_{\text{train}} \approx 1$–$2\times$), then degrades.

Implication. Law selection on the data axis is sensitive to the number of evaluation points. The qualitative finding—data scaling has bounded diminishing returns—is robust, but the specific functional form is not identifiable from typical experimental grids.

10.2 Resolution and Capacity Axes

On the resolution axis (4 points), logarithmic wins for most MLP/DeepONet slices, linear wins for FNO on Burgers. On the widened 10-point grid, FNO clearly shifts to linear (81% of slices), while MLP and DeepONet retain logarithmic as a narrow plurality. The capacity axis ($n = 7$ points) shows the most stable law selection, likely because it has the most data points per slice.

10.3 Sensitivity to Grid Density

The shift in data-axis law selection from logarithmic (5 points) to bounded-decay (11 points) is an important methodological warning: *scaling-law identification requires dense grids.* At $n = 4$–5 points per slice, even AICc (designed for small samples) cannot reliably distinguish functional forms. We recommend a minimum of 8–10 evaluation points per axis for any future scaling study that aims to identify the functional form, not merely the direction.

11 Cross-Resolution Transfer

For resolution-invariant architectures (FNO, DeepONet), we evaluate models trained at R_{train} on inputs at $R_{\text{eval}} \neq R_{\text{train}}$.

FNO on Burgers performs best at $R_{\text{eval}}/R_{\text{train}} \approx 1$–$2\times$, with test error of $\sim$0.25. At $16\times$ higher evaluation resolution, error returns to 0.43—near untrained levels. The model's learned spectral filters provide modest upward transfer but not dramatic zero-shot super-resolution.

FNO on Darcy is essentially flat across all ratios ($\varepsilon_{\text{rel}} \approx 0.68$–$0.70$): the task has such weak resolution dependence that cross-resolution evaluation adds no information.

DeepONet on Darcy degrades monotonically from $\varepsilon_{\text{rel}} = 1.36$ (ratio = 0.12) to 1.57 (ratio = 16), suggesting the architecture encodes resolution-specific features in its branch network.

An ablation comparing interpolation-based and subsampling-based cross-resolution evaluation (Appendix D) confirms that these findings are not driven by interpolation artifacts.

12 Failure Modes

12.1 Darcy Flow: A Pathological Case

Darcy flow stands out as anomalous across every analysis dimension:

- **High error floor.** The best achievable error is $\varepsilon_{\text{rel}} = 0.471$ (FNO). No model reaches below 47% relative error—compared to 2.7% (Burgers) and 0.2% (diffusion).
- **Extreme variance.** DeepONet on Darcy has a coefficient of variation (CV) of up to 1.54 across seeds. Mean error values of $\varepsilon_{\text{rel}} > 1$ (worse than predicting the mean) appear frequently.
- **Pervasive non-monotonicity.** Dozens of configurations show *increasing* error when data, resolution, or capacity is added. DeepONet at multiple capacity levels shows ε_{rel} jumping from 0.80 to 1.88 when going from $N = 1000$ to $N = 2000$.
- **Low explanatory power.** The log-linear regression R^2 for DeepONet on Darcy is only 0.35: none of the three scaling axes adequately explains the variance.

Interpretation. The discontinuous permeability coefficients create a fundamentally harder approximation problem. The highly nonlinear map from discontinuous coefficients to smooth solutions may require much larger datasets, architectures with attention mechanisms, or task-specific preprocessing (e.g., log-transformed coefficients). All scaling conclusions for Darcy should be interpreted cautiously.

12.2 DeepONet Saturation

DeepONet exhibits two distinct failure modes:

Early saturation. On Burgers, DeepONet's data-scaling exponent ($\alpha_N \approx -0.04$ to -0.07 at $R \geq 32$) is an order of magnitude weaker than FNO's. Even with a 100× increase in data ($N = 50 \to 5000$), error drops only ∼25%, compared to FNO's ∼85% reduction. On diffusion, DeepONet plateaus at $\varepsilon_{\text{rel}} \sim 0.10$–$0.22$ regardless of data, while FNO reaches 0.002.

More data can hurt. On Burgers, DeepONet with large/xlarge capacity consistently shows 5–15% error *increases* when going from $N = 2000$ to $N = 5000$, across multiple resolutions. This may reflect the branch-trunk decomposition reaching expressiveness saturation before the training set is fully utilized, leading to degraded optimization at larger N.

We do not claim DeepONet is fundamentally limited. The fixed training protocol may disadvantage it, and the architecture has shown strong results in other settings [9]. Within our controlled protocol, however, the saturation is systematic and reproducible.

13 Reproducibility and Noise

All configurations are replicated across 9 seeds (3 data × 3 training). Table 10 reports the coefficient of variation (CV $= \sigma/\mu$) across seeds:

Darcy is extremely noisy for non-FNO architectures (CV = 0.25), meaning a typical run's error varies by ±25% of the mean from seed changes alone. The noisiest individual configuration

Table 10: Mean coefficient of variation (CV) across seeds per task–model pair. Darcy is 3–8× noisier than other tasks.

Task	**FNO**	**MLP baseline**	**DeepONet**
Burgers	0.08	0.03	0.05
Darcy	0.07	0.25	0.25
Diffusion	0.17	0.07	0.11

(DeepONet/med-large/$N = 2000/R = 256$ on Darcy, CV = 1.54) produces runs ranging from near-zero to several times the mean error. This noise level severely limits the reliability of single-seed evaluations and underscores the importance of the multi-seed design.

14 Robustness and Ablations

14.1 Optimizer Sensitivity

All main results use Adam with learning rate 10^{-3} and patience 200. We re-run Burgers (all three models, all capacities) under two alternative configurations:

- **Variant A:** Adam with 2× patience (400 epochs)
- **Variant B:** SGD with momentum 0.9 + cosine annealing

for a total of 1,413 ablation runs.

Resolution axis: Law selection is fully stable. Logarithmic wins in 88% of baseline slices, 90% for Adam 2×, and 100% for SGD cosine. This is the strongest invariance result in the study.

Data axis: The specific law identity is more sensitive (DeepONet shifts from power to exponential; MLP shifts from saturation to power), but all winners belong to bounded-decay families. The qualitative conclusion—data scaling is bounded, not unbounded—is optimizer-invariant.

Conclusion: Directional findings are robust to the training configuration. The specific functional form on the data axis is training-sensitive, consistent with the grid-density sensitivity discussed in Section 10.3.

14.2 Densified Data Grid

Extending the data axis from 5 to 11 values ($N \in \{50, \ldots, 5000\}$ with 6 intermediate points) produces two systematic changes: (1) logarithmic is displaced as the dominant winner in favor of bounded-decay families (exponential, saturation); (2) AICc-BIC concordance improves from ∼10% to >50%. Both are consistent with the denser grid resolving the curvature near the error floor that was ambiguous at 5 points.

14.3 Widened Resolution Grid

Extending the Burgers resolution grid from 4 to 10 values ($R \in \{16, \ldots, 512\}$) reveals that FNO resolution scaling is clearly linear (81% of slices), while MLP and DeepONet retain logarithmic as a narrow plurality with significant linear competition. The directional finding—resolution helps for Burgers—is stable regardless of grid density.

15 Discussion

15.1 Summary of Evidence

Table 11 classifies findings by evidence strength.

Table 11: Summary of findings and evidence strength.

Finding	**Strength**	**Practical implication**
FNO dominates all comparisons (143/143)	Very strong	Use FNO for 1D operator learning
Dataset size is the primary lever (65–93%)	Strong	Prioritize data collection over model tuning
Resolution hurts on smooth problems	Strong	Use the lowest R that captures the physics
Capacity has weak returns ($\alpha_D \approx 0$)	Strong	A modest model suffices
No universal scaling law form	Moderate	Do not assume power law; validate per task
DeepONet saturates early	Strong	Consider alternatives for data-rich regimes
Darcy is not well-served by any model	Strong	Requires architectural innovation
Cross-resolution transfer is modest	Moderate	Do not rely on low-R training for high-R deployment

15.2 Resolution as Physics vs. Resolution as Dimensionality

The resolution axis conflates two distinct effects that are often treated as one:

1. **Resolution as physics information.** Higher resolution captures finer spatial features (shocks, discontinuities). For Burgers, where the solution develops sharp gradients, additional grid points carry genuinely new information.
2. **Resolution as input dimensionality.** Higher resolution increases the size of the input/output vectors. For diffusion, where the solution is spectrally sparse, additional grid points beyond $R \approx 32$ carry no new information but increase the dimensionality of the optimization problem.

Our results show that the second effect can dominate: when the additional grid points carry noise or irrelevant high-frequency content, models (especially small models) are better off at lower resolution. This has direct implications for benchmark design: reporting operator-learning accuracy only at high resolution conflates learning ability with the model's tolerance for input dimensionality.

15.3 Implications for Practice

For practitioners working with 1D PDEs similar to our benchmarks:

1. **Architecture first.** The choice of inductive bias (spectral vs. pointwise vs. MLP) has a larger effect than any amount of data, resolution, or capacity scaling.
2. **Data second.** Collecting ∼500–1000 high-quality samples provides most of the achievable accuracy. Returns diminish sharply beyond this range.

3. **Resolution carefully.** Match resolution to the problem's spectral content, not to the solver's capability. Using the highest available resolution can actively hurt.
4. **Capacity modestly.** A medium-sized model (∼30–50K parameters) captures most of the available accuracy for these 1D tasks; scaling to millions of parameters provides negligible returns.

15.4 Comparison with NLP Scaling Laws

The NLP scaling regime [5, 4] features three properties that our 1D PDE setting does not share: (1) power-law relationships spanning many orders of magnitude in model size; (2) weak diminishing returns across the entire tested range; (3) the same functional form (power law) governing all axes. In our setting, (1) the parameter range is modest (∼2K–2.7M), (2) diminishing returns set in by $N \approx 1000$ and $D \approx$ small, and (3) the functional form is regime-dependent. This suggests that the smooth, universal scaling regime observed in NLP may require the problem complexity and model scale to be in an appropriate relationship—conditions not met by simple 1D PDEs with sub-million-parameter models.

15.5 Limitations

1. **1D only.** All tasks are 1D PDEs. Resolution may be far more important in 2D/3D, where discretization error compounds with dimensionality. We make no claims about higher-dimensional settings.
2. **Fixed training protocol.** A shared optimizer and schedule ensures controlled comparison but may disadvantage specific architectures (e.g., DeepONet may benefit from different learning-rate schedules). The optimizer ablation partially addresses this but does not exhaust the hyperparameter space.
3. **Limited resolution range for some tasks.** Darcy uses only 4 resolution values in the baseline grid. The wide-R extension is complete only for Burgers.
4. **Relative L^2 only.** Different error norms (pointwise max, Sobolev, spectral) could change accuracy rankings, especially for Burgers (localized shocks).
5. **Canonical PDEs.** Burgers, Darcy, and diffusion are standard benchmarks, not representative of the full complexity of scientific applications. Scaling behavior on turbulence, multi-physics, or real-world engineering problems may differ substantially.

16 Conclusion

We have presented a controlled empirical study of scaling laws for neural operator learning across three canonical 1D PDE tasks, three architectures, and two orders of magnitude along dataset size, resolution, and model capacity, totaling 32,783 training runs.

The study yields three central messages. First, for 1D problems with periodic or simple boundary conditions, architecture choice (specifically, FNO's spectral inductive bias) dominates all other scaling considerations by a wide margin. Second, dataset size is the only reliable scaling axis: resolution is task-dependent (helpful for shocks, harmful for smooth solutions, irrelevant for elliptic problems) and capacity returns are negligible past modest model sizes. Third, the functional form of the scaling relationship is not universal: it depends on the task, architecture, and regime, and shifts with the density of the evaluation grid.

These results suggest that the era of "scaling everything" in operator learning may not mirror NLP's experience, at least for 1D benchmarks: practitioners should prioritize choosing the right architecture and collecting sufficient training data over scaling model size or discretization resolution. The extent to which these conclusions transfer to higher-dimensional, multi-physics, or industrial-scale problems remains an important open question.

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A Capacity Grid

B PDE Solver Details

Burgers equation. $u_t + uu_x = \nu u_{xx}$ on $[0, 2\pi)$ with periodic boundary conditions, $\nu = 0.01$, $T = 1.0$. Solved with integrating-factor RK4 in Fourier space with $\frac{2}{3}$-rule dealiasing and $\Delta t = 10^{-3}$.

Table 12: MLP and DeepONet capacity grid. FNO uses a separate grid over modes, width, and number of layers.

Level	**Layers**	**Width**	$\sim D$ **(MLP,** R=128)
tiny	2	32	∼5K
small	2	64	∼13K
small-med	2	96	∼22K
medium	2	128	∼34K
med-large	2	192	∼62K
large	2	256	∼82K
xlarge	3	256	∼148K

Random initial conditions are superpositions of Fourier modes with random phases and amplitudes drawn from a k^{-2} energy spectrum.

Darcy flow. $-\nabla\cdot(a(x)\nabla u(x)) = f(x)$ on $[0, 1]$ with Dirichlet conditions, $f(x) = 1$. The coefficient field $a(x)$ is drawn from a log-normal random field. Solved via spectral Galerkin method.

Diffusion equation. $u_t = \nu u_{xx}$ on $[0, 2\pi)$ with periodic boundary conditions, $\nu = 0.01$, $T = 1.0$. Exact spectral solution: $\hat{u}_k(t) = \hat{u}_k(0)e^{-\nu k^2 t}$. Same initial-condition distribution as Burgers.

C Multi-Law Fitting Details

Six candidate laws are fit via `scipy.optimize.curve_fit` (Levenberg–Marquardt) with maximum 10,000 iterations:

Table 13: Candidate functional forms. X is the scaling variable (N, R, or D); E_∞ denotes the irreducible error floor.

Family	**Form**	**Parameters**
Power	$E_\infty + aX^{-\alpha}$	3
Exponential	$E_\infty + ae^{-\alpha X}$	3
Logarithmic	$a - b\ln X$	2
Stretched exp.	$E_\infty + ae^{-(\alpha X)^\beta}$	4
Saturation	$E_\infty + a/(X + b)$	3
Linear	$a + bX$	2

AICc is computed as $\text{AICc} = n\ln(\text{RSS}/n) + 2k + 2k(k + 1)/(n - k - 1)$, where n is the number of points per slice and k is the number of parameters. When $n \leq k + 1$, AICc is set to $+\infty$.

D DeepONet Cross-Resolution Ablation

We compared interpolation-based and subsampling-based cross-resolution evaluation across 288 comparisons (Darcy and Diffusion, 3 capacities $\times$ 4 training resolutions $\times$ 4–5 evaluation resolutions $\times$ 4 dataset sizes).

When $R_{\text{eval}} > R_{\text{train}}$ (upsampling): interpolation and subsampling perform comparably (mean ε_{rel}: 0.523 vs. 0.520), with subsampling winning 68% of 180 comparisons by a slim margin. When $R_{\text{eval}} < R_{\text{train}}$ (downsampling): interpolation is overwhelmingly better (98% of 108 cases; mean ε_{rel}: 0.531 vs. 1.068).

The standard interpolation strategy is robust: cross-resolution transfer conclusions are not driven by interpolation artifacts.

E BIC Concordance

Because AICc and BIC penalties diverge at small sample sizes (at $n = 5$, BIC's penalty $k \ln n \approx 1.6k$ is *weaker* than AICc's corrected penalty), low concordance at small n reflects the well-understood divergence between criteria, not evidence against the AICc-selected law.

Table 14: AICc–BIC concordance rate per axis.

Task	**Model**	**Data**	**Resolution**	**Capacity**
Burgers	MLP	4%	3%	90%
Burgers	DeepONet	21%	3%	85%
Burgers	FNO	11%	9%	70%
Darcy	MLP	4%	43%	80%
Darcy	DeepONet	11%	69%	95%
Darcy	FNO	14%	34%	65%
Diffusion	MLP	4%	97%	45%
Diffusion	DeepONet	11%	100%	80%
Diffusion	FNO	21%	74%	90%